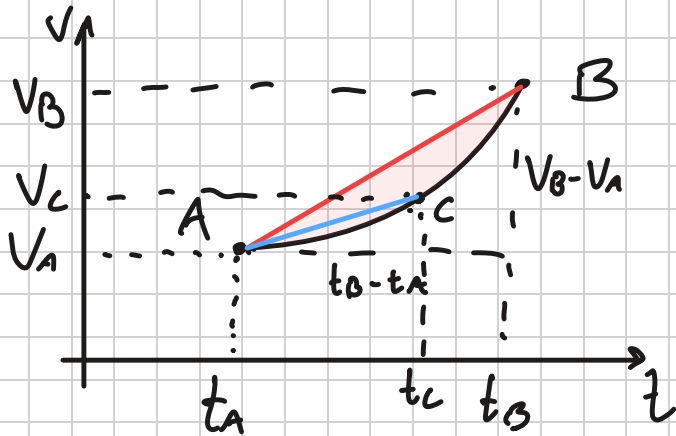


Week 2

instantaneous acceleration

$$a = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt} \quad \text{where } v(t)$$



on AB :

$$\underline{a_{ave}} = \frac{v_B - v_A}{t_B - t_A}$$

On AC : $a_{ave} = \frac{v_C - v_A}{t_C - t_A}$

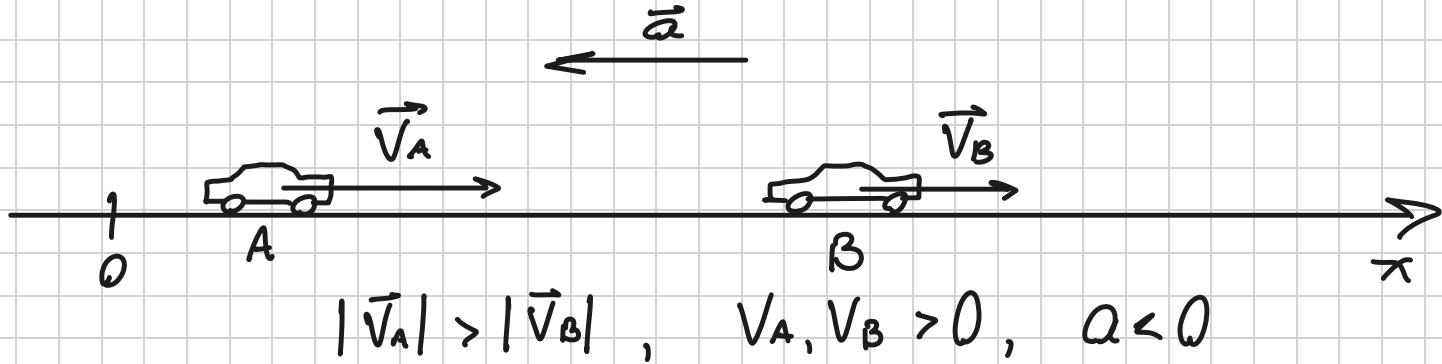
$$a_{ave AB} \neq a_{ave AC}$$

Let C slide toward A such that $t_C - t_A \rightarrow 0$

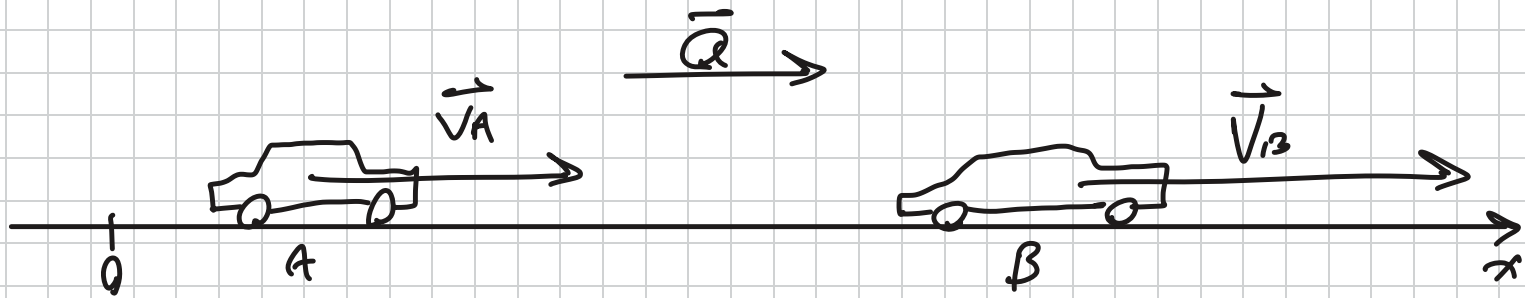
$$a_{inst} = \lim_{\Delta t \rightarrow 0} \frac{\Delta v}{\Delta t}$$

Note : a, v do not necessarily point in the same direction

1.

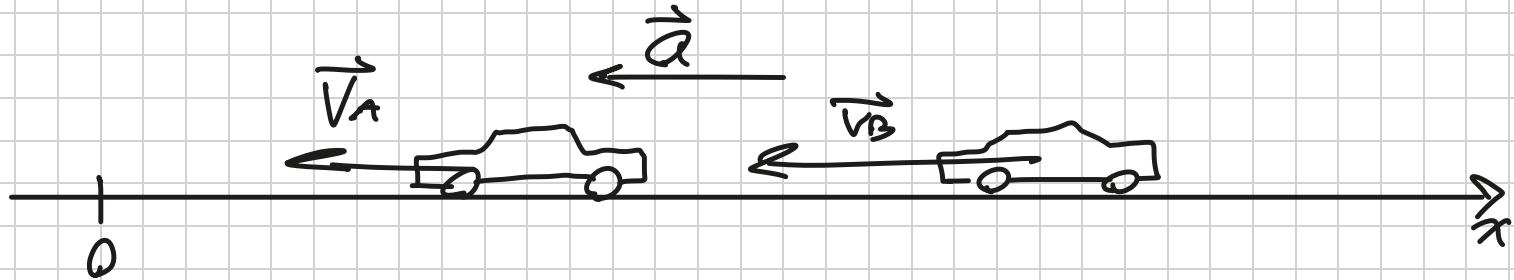


2.

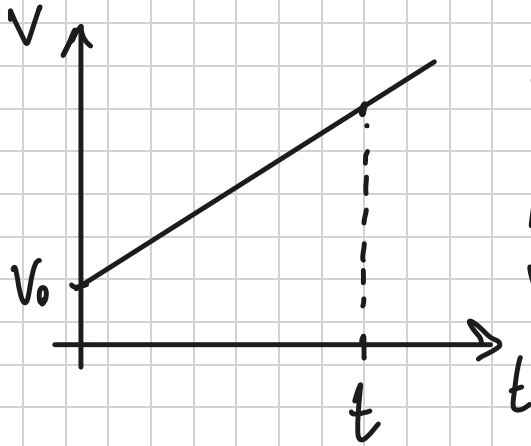


$$|\vec{v}_A| < |\vec{v}_B|, v_A, v_B > 0, Q > 0$$

Exercise: redo diagram. $\vec{v} < 0$



2.4 motion with $\vec{a}$ constant



$$v(t) = c + d \cdot t$$

$$|| v(0) = v_0$$

$$v(0) = c \Rightarrow c = v_0$$

$$\frac{dv}{dt} = d = a$$

$$\Rightarrow \boxed{v(t) = v_0 + at}$$

What is V_{ave} between D and t

$$x(0) = x_0 \quad V_{ave} = \frac{x - x_0}{t}$$

$$\Rightarrow x - x_0 = V_{ave} \cdot t$$

Because $V(t)$ is linear, $V_{ave} = \frac{V_0 + V(t)}{2}$

$$\Rightarrow V_{ave} = \frac{V_0 + V_0 + at}{2} = V_0 + \frac{at}{2}$$

$$\Rightarrow x - x_0 = \left(V_0 + \frac{at}{2}\right)t = V_0 t + \frac{at^2}{2}$$

$$\Rightarrow \boxed{x(t) = x_0 + V_0 t + \frac{1}{2}at^2}$$

$$\frac{dx}{dt} = \frac{d}{dt} \left(x_0 + V_0 t + \frac{1}{2}at^2 \right) = V_0 + at$$

Eliminate t

$$at = V - V_0 \Rightarrow t = \frac{V - V_0}{a}$$

$$\left[x = x_0 + V_0 \frac{V - V_0}{a} + \frac{1}{2} a \left(\frac{V - V_0}{a} \right)^2 \right] \cdot 2a$$

$$2a(x - x_0) = 2V_0(V - V_0) + (V - V_0)^2$$

$$2a(x - x_0) = \cancel{2V_0V} - 2V_0^2 + V^2 + V_0^2 - \cancel{2V_0V}$$

$$2a(x - x_0) = V^2 - V_0^2$$

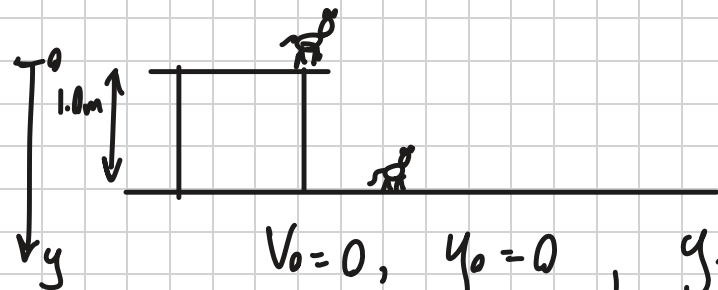
$$V^2 - V_0^2 = 2a \cdot (x - x_0)$$

Let $x - x_0 = d$

$$V^2 - V_0^2 = 2a \cdot d$$

example

Uniform downward $\vec{a} = 9.8 \text{ m/s}^2$



$$V_0 = 0, \quad y_0 = 0, \quad y_1 = 1.0 \text{ m}$$

what's the v of cat when it hits the floor

$$v^2 - V_0^2 = 2g \cdot d \Rightarrow v = \sqrt{2gd}$$

$$v = \sqrt{2 \times 9.8 \times 1.0} = \underline{4.4} \text{ m/s}$$

free fall

- gravity $\vec{a}$ is constant

if air resistance and buoyancy are neglected
 $\Rightarrow$ All objects fall the same acceleration of gravity

if we are close to the earth surface $\Rightarrow$ constant acceleration

$$g = 9.8 \text{ m/s}^2$$


$$a = -g \quad (\text{negative})$$

the 3 basic equations:

$$y = y_0 + v_0 t + \frac{1}{2} a t^2$$

$$y(0) = y_0, \quad v(0) = v_0$$

$$a = -g$$

$$y(t) = y_0 + v_0 t - \frac{1}{2} g t^2$$

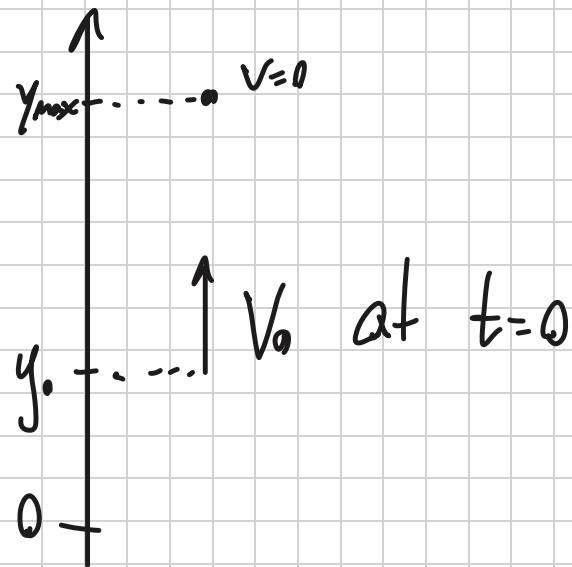
$$v(t) = v_0 + a t$$

$$v(t) = v_0 - g t$$

$$V^2 - V_0^2 = 2a(y - y_0)$$

$$\Rightarrow V^2 = V_0^2 - 2g(y - y_0)$$

from position y_0 with initial velocity of V_0



object will reach
a max height, then
fall back down
use equation (3), $v=0$

$$0 = V_0^2 - 2g(y_{\max} - y_0)$$

$$\Rightarrow y_{\max} - y_0 = \frac{V_0^2}{2g} \Rightarrow y_{\max} = y_0 + \frac{V_0^2}{2g}$$

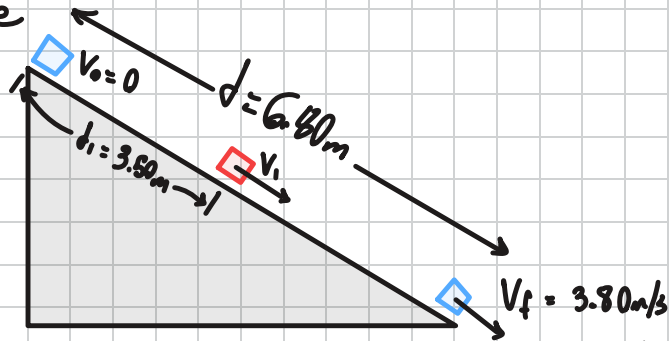
position 2 same point where it was
launched, but now on the way down

$$y_2 = y_0 : v_2^2 = v_0^2 - 2g(\cancel{y_2 - y_0})$$

$$\Rightarrow v_2^2 = v_0^2 \Rightarrow |v_2| = |v_0|$$

$$v_2 = -v_0$$

Example



What is the velocity halfway down the incline

① find the $\vec{a}$:

$$x_0 = 0, \quad x_f = 6.80\text{m}$$

$$v_0 = 0, \quad v_f = 3.80\text{m/s}$$

$$v_f^2 = v_0^2 + 2a(x_f - x_0) = 2ad$$

$$\Rightarrow a = \frac{v_f^2}{2d} = 1.06176\text{m/s}^2$$

② Find V_i :

$$V_i^2 = 2ad_i$$

$$\Rightarrow V_i = \sqrt{2ad_i}$$

OR: $V_i = \sqrt{2 \frac{V_f^2}{d} d_i} \Rightarrow V_i = \sqrt{\frac{d_i}{d}} \cdot V_f$

$$V_i = 2.69 \text{ m/s}$$

Example 2

Bobby the flea can jump 0.440m

a) what's his initial velocity?

$$V_0^2 = 2gh_{\max} \Rightarrow V_0 = \sqrt{2gh_{\max}}$$

$$g = 9.8 \text{ m/s}^2 \quad \text{use the lowest sig fig}$$

$$V_0 = 2.9 \text{ m/s} \quad (\text{two sig figs})$$

b) what is bobby total airtime?

V_0 is upward at $t=0$, $y=0$

V_2 is velocity at $y=0$ on the way down

$$(V_2 = -V_0)$$

$$V_2 = V_0 + at$$

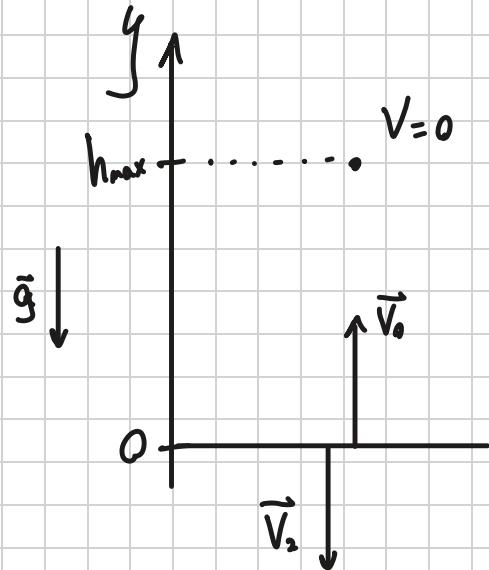
$$= V_0 - gt$$

$$gt = V_0 - V_2$$

$$= 2V_0$$

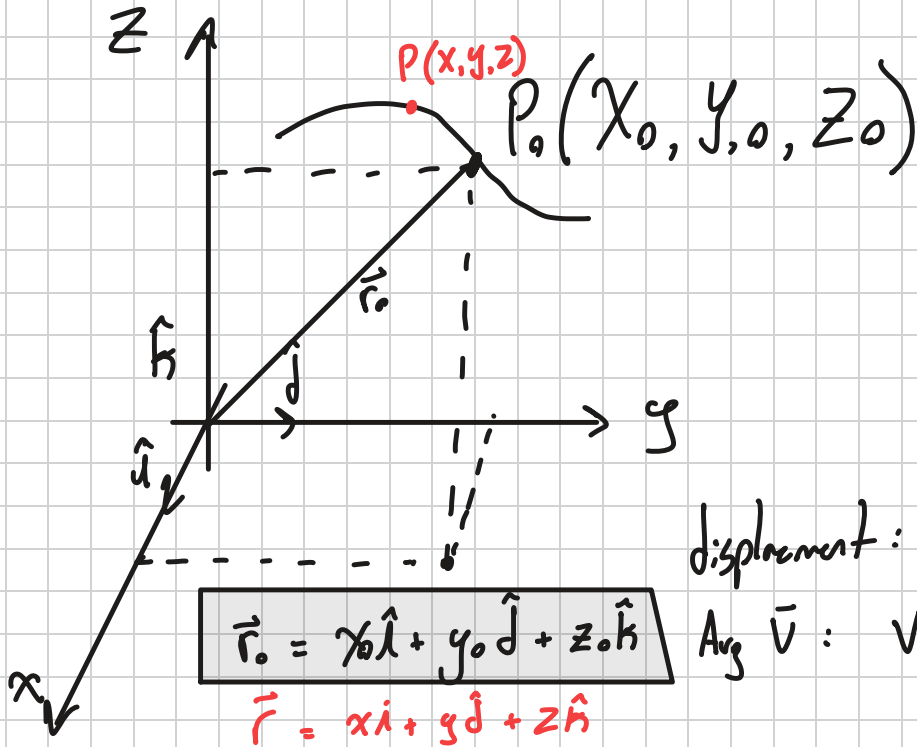
$$\Rightarrow t = \frac{2V_0}{g} = \frac{2\sqrt{2gh}}{g} = 2\sqrt{\frac{2h}{g}} = \text{Answer}$$

I'm not calculating this



Chapter 3

=> position, Velocity Vectors



$\hat{i}, \hat{j}, \hat{k}$ unit vectors
in the x, y, z

displacement: $\vec{r} - \vec{r}_0 = (x - x_0) \hat{i} + (y - y_0) \hat{j} + (z - z_0) \hat{k}$

Avg $\vec{v}$: $v_{ave} = \frac{\Delta \vec{r}}{\Delta t} = \frac{\Delta x \hat{i} + \Delta y \hat{j} + \Delta z \hat{k}}{\Delta t} = \frac{\Delta x}{\Delta t} \hat{i} + \frac{\Delta y}{\Delta t} \hat{j} + \frac{\Delta z}{\Delta t} \hat{k}$

instantaneous velocity

$$\vec{V} = \lim_{t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt} = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k}$$

$$\vec{V} = V_x \hat{i} + V_y \hat{j} + V_z \hat{k}, \quad V_x = \frac{dx}{dt}, \quad V_y = \frac{dy}{dt}, \quad V_z = \frac{dz}{dt}$$

Example $\vec{r}(t) = (5t+17)\hat{i} + t^2\hat{j} - 3t\hat{k}$

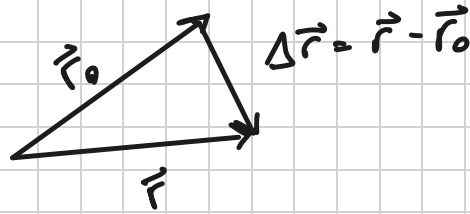
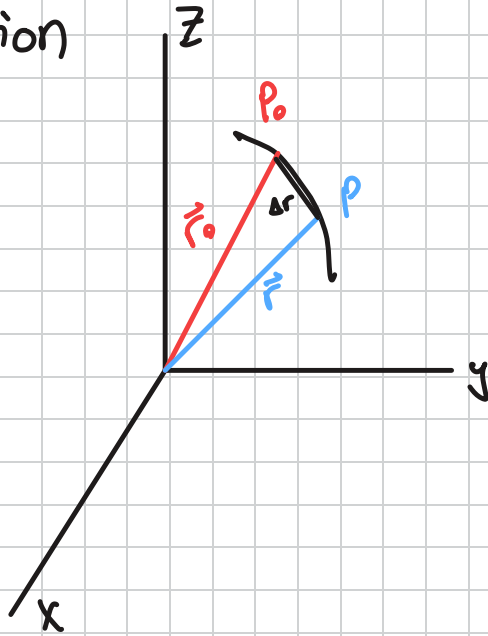
Calculate $\vec{v}(t)$, and the speed v

$$\vec{v} = \frac{d\vec{r}}{dt} = 5\hat{i} + 2t\hat{j} - 3\hat{k}$$

Speed $v = |\vec{v}| = \sqrt{v_x^2 + v_y^2 + v_z^2}$

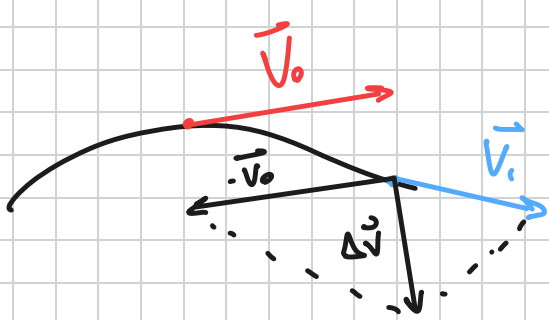
$$v(t) = \sqrt{25 + 4t^2 + 9} = \sqrt{34 + 4t^2}$$

Acceleration



$$\vec{v}_{ave} = \frac{\Delta \vec{r}}{\Delta t}, \quad \vec{v} = \lim_{t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

Note: $\vec{v}$ is always tangent to the trajectory



$$\vec{a}_{avg} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_1 - \vec{v}_0}{\Delta t}$$

$$\Delta \vec{v} = \vec{v}_1 - \vec{v}_2 = \vec{v}_1 + (-\vec{v}_0)$$

$$\vec{a}_{avg} \parallel \Delta \vec{v}$$

Note: $\vec{a}$ always points to the inside of trajectory
(if it's a curve)

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} + \frac{dv_z}{dt} \hat{k}$$



$$\vec{a} = a_x \hat{i} + a_y \hat{j} + a_z \hat{k}, a_x = \frac{dv_x}{dt}, a_y = \frac{dv_y}{dt}, a_z = \frac{dv_z}{dt}$$

$$\text{or: } \vec{a} = \frac{d^2 \vec{r}}{dt^2}, a_x = \frac{d^2 x}{dt^2}, a_y = \frac{d^2 y}{dt^2}, a_z = \frac{d^2 z}{dt^2}$$

Any particle moving on a curved path has non-zero acceleration, because the velocity vector is changing

parallel, perpendicular acceleration

$$\vec{a} = \vec{a}_{||} + \vec{a}_{\perp} \quad (\text{relative to the trajectory})$$

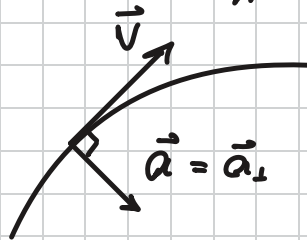
$\vec{a}_{||}$ - measures the rate of change of the speed $|\vec{v}|$

$\vec{a}_{\perp}$ - measures the rate of change of the direction of $\vec{v}$

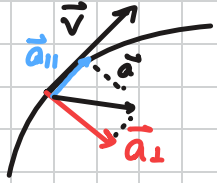
Example

① motion on a curve with constant speed

$$\vec{a}_{||} = 0, \quad \vec{a} = \vec{a}_{\perp}$$



② motion with increasing speed



$\vec{a}$ points inside and forwards